MILITARY SPECIFICATION

PACKAGING OF RATION, COLD WEATHER COMPONENTS (IN POUCHES)

This specification is approved for use by Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers packaging of the Ration, Cold Weather component items.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

L-C-110	-	Cellophane (Coated and Noncoated Regenerated Cellulose Film)
L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
QQ-A-1876	-	Aluminum Foil
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-843	-	Cushioning Material, Cellulosic

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 89GP

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PPP-C-1752 - Cushioning Material, Unicellular Polyethylene Foam Flexible (For Packaging Purposes)
PPP-C-1797 - Cushioning Material, Resilient, Low Density, Unicellular Polypropylene Foam

MILITARY

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items

STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1238 - Flow Rates of Thermoplastics by Extrusion Plastometer
D 1505 - Density of Plastics by Density Gradient Technique
F 88 - Seal Strength of Flexible Barrier Material

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Components and materials.

3.1.1 Food product. The food product to be unit packed shall be as specified in the applicable component document (see 6.2).

3.1.2 Mixing bags. Mixing bags for the entree bars and instant oatmeal shall be fabricated from clear, high density 0.941 to 0.965 g/cm^3 food grade, flat polyethylene tubing. The bottom seal shall be of adequate strength and integrity to withstand the addition of boiling water when tested in accordance with 4.3.1.3. The size of the mixing bags shall be 7-1/2 inches wide by 8 inches long (inside dimensions) by 0.004 inch thick.

3.1.3 Inner bags (for oatmeal cookie and granola bars). To protect barrier pouches from the coarse finish of the product, the bars shall be inserted in unsealed (a) polyethylene bags or (b) wrapped in close fitting cellophane film:

a. Bags shall be fabricated from clear, food grade, linear low density polyethylene (0.915 to 0.935 g/cm^3) tubing sealed on one end. The dimensions shall be 2-1/2 inches wide by 4-1/2 inches long (inside dimensions) $\pm$ 1/16 inch by 0.002 inch thick.

b. As an alternative to a. above, each bar shall be wrapped and heat sealed in a sheet of 0.0014 inch thick cellophane film complying with type IV, class 2 (low water vapor permeability) of L-C-110. The wrapper shall be close forming and shall completely cover the product. The finished package shall be free of excess air or vented just prior to vacuum sealing in barrier pouches.

3.1.4 Mixing bag closure device. The mixing bag closure device shall be the Dennison "Flip-Clip" (see 6.3) or equivalent and shall be fabricated from molded white nylon. It shall be capable of keeping the mixing bag closed after the bag has been filled with the required amount of water, after the top of the bag has been twisted shut, and during the time required for rehydration of the product.

3.1.5 Barrier pouches.

3.1.5.1 Barrier pouches shall be fabricated from material specified in 3.1.5.1.1 or 3.1.5.1.2 and in table II.

3.1.5.1.1 Pouch material "A". The pouch shall be constructed from a multiple ply flexible laminate barrier material consisting of an outer ply of 0.0009 inch thick oriented polypropylene bonded to an intermediate ply of 0.0007 inch thick aluminum foil with 10 pounds per ream pigmented polyethylene, and bonding the opposite side of the aluminum foil to 0.003 inch thick ionomer (inner ply) or a blend of not less than 50 percent linear low density polyethylene and polyethylene. The linear low density polyethylene portion of the blend shall be the copolymer of ethylene and octene-1, having a melt index range of 0.8 to 1.2 g/10 minutes in accordance with ASTM D 1238 and a density range of 0.918 to 0.922 g/cm^3 in accordance with ASTM D 1505. The pigmented polyethylene shall be colored white overall with a color in the range of 37778 through 37886 of

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FED-STD-595. The material shall be suitably formulated for food packaging and shall not impart an odor or flavor to the product being packaged. The material shall show no evidence of delamination, degradation, or foreign odor when heat sealed or fabricated into pouches.

3.1.5.1.2 Pouch material "B". The pouch shall be constructed of a multiple ply flexible laminate barrier material consisting of an outer ply of 50 gage polyethylene terephthalate bonded to an intermediate ply or 0.00035 inch thick aluminum foil with 10 pounds per ream pigmented polyethylene, and bonding the opposite side of the aluminum foil to 0.003 inch thick (inner ply) blend of not less than 50 percent linear low density polyethylene and polyethylene. The linear low density polyethylene portion of the blend shall be the copolymer of ethylene and octene-1, having a melt index range 0.8 to 1.2 g/10 minutes in accordance with ASTM D 1238 and a density range of 0.918 to 0.922 g/cm^3 in accordance with ASTM D 1505. As an alternative to the blend of linear low density polyethylene and polyethylene, an ionomer may be used. Pouch color shall be as indicated in 3.1.5.1.1.

3.1.5.2 Pouch construction. The pouch shall be a flat style preformed pouch having inside dimensions as specified in table I. The first dimension specified in table I shall be the opening of the pouch between the heat sealed sides. The empty pouch shall be made by heat sealing three edges with 3/8 inch (± 1/8 inch) wide seals. The pouch shall maintain its integrity and air tightness of the side and bottom seals when tested as specified in 4.3.1.6. The side and bottom seals shall have an average seal strength of not less than 6 pounds per inch and no individual specimen shall have a seal strength of less than 5 pounds per inch when tested as specified in 4.3.1.5. A "V", "C", or block "U" tear notch at least 1/32 inch deep, located 1 to 1-1/4 inches from the top edge of the pouch, shall be made in one or both side seals. The distance between the inside edge of the tear notch and the inside edge of the seal shall be at least 3/16 inch. One side of the open end of the pouch shall be provided with an extended or foldover lip, extended not more than 1/8 inch (± 1/16 inch) to facilitate opening and filling. Tear notch location shall be measured from the top of the pouch, excluding the extended or foldover lip. The pouch shall be constructed with square or rounded corners at all four corners (radius approximately 3/8-inch).

TABLE I. Pouch dimensions

Item	Pouch inside dimensions (± 1/16 inch)
Entree (4 compressed bars - 4 formed slices)	5 by 7-1/2
Entree (5 or 6 formed slices)	5 by 9
Oatmeal cookie and granola bar	3-7/8 by 5-3/4
Beverage base, orange, compressed (2 single or 1 double)	2-3/4 by 5-1/2
Chocolate bars with toffee	2-3/8 by 5-1/2
Oatmeal cereal (3 flavors)	4-5/8 by 7-1/4

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TABLE I. Pouch dimensions (cont'd)

Item	Pouch inside dimensions (± 1/16 inch)
Lemon tea	2 by 4-1/2
Instant soup (chicken noodle)	3 by 5
Cocoa beverage powder	3-1/2 by 5
Fig or blueberry bars	2-3/4 by 5-5/8
Cookie, chocolate covered	3-3/4 by 5-1/4
Brownie, chocolate covered	3-3/4 by 5-1/4
Nut raisin mix	3-3/8 by 6
Apple cider	3 by 5-5/8
Fruit soup	3 by 5-5/8

3.2 Pouch filling and sealing. The ration components shall be inserted in pouches, as specified in the applicable component specification or the assembly specification, in such a manner as to avoid puncturing the pouch material or contaminating the heat seal area. The filled pouches shall be sealed in accordance with vacuum requirements specified in table II with a 3/8 inch (± 1/8 inch) heat seal. The sealed pouch shall show no loss of vacuum when examined as specified in 4.3.2. If thermal impulse closure seals are used, a seal width of 1/8 inch will be acceptable. The closure seal shall not extend below the tear notch on either side of the pouch. The closure seal shall be free of fold over wrinkles and delaminations. The closure seal shall have an average seal strength of not less than 6 pounds per inch and no individual specimen shall have a seal strength of less than 5 pounds per inch when tested as specified in 4.3.4

TABLE II. Vacuum and pouch material requirements

Item	Vacuum requirement (Hg)	Pouch material
Entree bars or slices	28	A
Oatmeal cookie and granola bars	28	A
Beverage base, orange, compressed bar(s)	28	A
Chocolate bars with toffee	22-24	A
Instant oatmeal	1/	B
Lemon tea	1/	B
Instant soup (chicken noodle)	2/	B
Cocoa beverage powder	1/	B
Fig or blueberry bars	8-12	B
Cookie, chocolate covered	22-24	A
Brownie, chocolate covered	22-24	A
Nut raisin mix	2/	A
Apple cider	1/	B
Fruit soup	1/	B

1/ Manually evacuate excess air before sealing. Be careful not to contaminate heat seal area with product.

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2/ Nitrogen flush before sealing.

3.3 Labeling and marking.

3.3.1 Mixing bags. Each mixing bag shall be clearly printed, in a manner that does not change the mixing bag material, with a permanent, grease/oil resistant, food compatible, black ink which is free of carcinogenic elements or ingredients. Lettering shall be 5/32 inch block letters. The following information, as applicable, shall be located on the body of the mixing bag:

a. Entree bag:

Break into quarters. Add a minimum of 1/2 canteen cup of hot water. Seal bag with plastic closure device. Wait 5 minutes before eating. Hot water may be added in stages to keep food hot.

b. Oatmeal bag:

Add a minimum of 1/3 canteen of hot water. Seal bag with plastic closure device. Wait one minute before eating. Optional: if desired, crumbled granola bars, fruit soup, or apple cider may be added for variety.

3.3.2 Inner bags: (for oatmeal cookie and granola bars). Each inner bag shall be clearly printed or stamped with the name of the product, in a manner that does not damage the individual bag material, with a permanent, grease/oil resistant, food compatible, black ink which is free of carcinogenic elements or ingredients. Lettering shall be 5/32 inch block letters.

3.3.3 Pouches. Each pouch shall be clearly printed or stamped, in a manner that does not damage the pouch, with permanent black ink which is free of carcinogenic elements or ingredients. The following information shall be located on the body of the pouch not closer than 1/16 inch to any seal. All pouches shall be coded. If a non-contact type printer is used, the following information, as applicable, may be located anywhere on the pouch (in one complete print), except the closure seal area. Note: In all cases, it is the contractor's responsibility to independently receive label approval from the USDA. The label shall reflect contractor's name, address, number, date markings, USDA stamp, ingredients, and weight for all products.

a. Entree bars or slices:

Product name (letters not less than 7/32 inch high)
MIXING BAG WITH INSTRUCTIONS INSIDE (letters not less than 5/32 inch high)
Ingredients (letters not less than 3/32 inch high)
Date markings (in accordance with MIL-L-1497) (letters not less than 1/4 inch high)

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Net weight (letters not less than 1/8 inch high)
Official establishment number (e.g. EST-38) (letters not less than 1/8 inch high)
Name and address of producer (letters not less than 1/8 inch high)

b. Oatmeal cookie and granola bars:

Product name (letters not less than 7/32 inch high)
Name and address of producer (letters not less than 1/8 inch high)

c. Orange beverage bar/s: (Product name not less than 7/32 inch high)

EAT AS IS OR BREAK INTO SMALL PIECES IN CANTEEN CUP. ADD 1/3 CANTEEN CUP OF WATER. STIR UNTIL DISSOLVED. (Directions not less than 1/8 inch high)
Name and address of producer (letters not less than 3/32 inch high)

d. Chocolate bars with toffee:

CANDY, CHOCOLATE BARS WITH TOFFEE, ENRICHED SWEET (letters not less than 3/32 inch high)
Name and address of producer (letters not less than 1/8 inch high)

e. Instant oatmeal: (Product name not less than 7/32 inch high)

MIXING BAG WITH INSTRUCTIONS INSIDE (letters not less than 1/8 inch high)
Name and address of producer (letters not less than 3/32 inch high)

f. Tea, instant, lemon flavored: (Product name not less than 7/32 inch high)

ADD 1/3 CANTEEN CUP HOT OR COLD WATER TO CONTENTS AND STIR (letters not less than 3/32 inch high)
Name and address of producer (letters not less than 3/32 inch high)

g. Soup, instant, chicken noodle: (Product name not less than 7/32 inch high)
ADD APPROXIMATELY 1/3 CANTEEN CUP HOT WATER TO CONTENTS AND STIR.
Wait 1-2 minutes. If desired, used packaging material can be wrapped around canteen cup to keep contents warm (directions not less than 1/8 inch high)
Name and address of producer (letters not less than 3/32 inch high)

h. Cocoa beverage powder: Type I, fortified 1-1/2 oz. net (Product name not less than 1/8 inch high)
MIX CONTENTS WITH 1/4 CANTEEN CUP WATER
FOR HOT COCOA, ADD CONTENTS TO HOT WATER AND STIR UNTIL DISSOLVED.
FOR COLD COCOA, MIX CONTENTS WITH ABOUT 1 FLUID OUNCE OF WATER TO MAKE A SMOOTH PASTE AND THEN ADD REMAINDER OF COLD WATER AND STIR UNTIL BLENDED. (Directions not less than 1/16 inch high)
Name and address of producer (letters not less than 3/32 inch high)

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i. Fig bars/blueberry bars: (Product name not less than 7/32 inch high) 1/
Name and address of producer (letters not less than 3/32 inch high)

j. Cookie, chocolate covered: (Product name not less than 7/16 inch high)
Name and address of producer (letters not less than 3/32 inch high)

k. Brownie, chocolate covered: (Product name not less than 7/16 inch high)
Name and address of producer (letters not less than 3/32 inch high)

l. Nut raisin mix: (Product name not less than 7/32 inch high)
Name and address of producer (letters not less than 3/32 inch high)

m. Apple cider: (Product name not less than 7/32 inch high)

Instructions: ADD 1/2 CANTEEN CUP OF HOT OR COLD WATER TO CONTENTS AND STIR. (letters not less than 5/32 inch high)
Name and address of producer (letters not less than 3/32 inch high)

n. Fruit soup: (Product name not less than 7/32 inch high)

Instructions: ADD 1/2 CANTEEN CUP OF HOT OR COLD WATER TO CONTENTS AND STIR. (letters not less than 5/32 inch high)
Name and address of producer (letters not less than 3/32 inch high)

1/ The fig and blueberry bars may be packaged in a barrier pouch in an original vendor package. If ingredient marking and name and address of producer is missing, it shall be added to barrier pouch.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. USDA will determine the degree of inspection necessary to assure compliance with the requirements of this document.

4.3 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

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4.3.1.1 Material certification. Material listed below shall be accepted on the basis of a contractor's certificate of conformance to the indicated requirements.

Requirement	Requirement paragraph	Test procedure
Mixing bag and inner bag material identification	3.1.2 and 3.1.3	Laboratory evaluation
Mixing bag closure device	3.1.4	Visual examination
Thickness of films for laminated material	3.1.5.1.1 and 3.1.5.1.2	As specified in L-P-378, except that a machinists' micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378
Aluminum foil thickness	3.1.5.1.1 and 3.1.5.1.2	As specified in QQ-A-1876
Laminated material identification and construction	3.1.5.1.1 and 3.1.5.1.2	Laboratory evaluation
Color of laminated material	3.1.5.1.1 and 3.1.5.1.2	Visual evaluation

4.3.1.2 Unfilled mixing bag or inner bag examination as applicable. The unfilled mixing bags or inner bags shall be examined for the defects listed in table III. The lot size shall be expressed in bags. The sample unit shall be one unfilled mixing bag or inner bag. The inspection level shall be S-4 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 0.65 for major defects.

TABLE III. Unfilled mixing bag or inner bag defects

Category	Defect
Major	
101	Not formed as specified
102	Inside dimensions not as specified
103	Tear, hole, or open seal
104	Required marking missing, incorrect, or illegible
105	Not material specified
106	Bags stick together
107	Not clean
108	Foreign odor

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4.3.1.3 Unfilled mixing bag leakage testing. The unfilled mixing bags shall be tested for leakage as follows: Each sample bag shall be filled with 8 ounces of boiling water (being careful not to spill any water on the outside) and suspended 8 to 12 inches above a sheet of clean paper for 10 minutes. Any moisture on the paper attributable to bag leakage shall be considered a major defect. The lot size shall be expressed in bags. The sample unit shall be one bag. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.3.1.4 Unfilled pouch examination. The unfilled pouches shall be examined for the defects listed in table IV. The lot size shall be expressed in pouches. The sample unit shall be one unfilled pouch. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects and 4.0 for minor defects.

TABLE IV. Unfilled pouch defects

Category		Defects
Major	Minor	
101		Not size specified
102		Heat seal dimensions on three edges not as specified
103		Tear, hole, or open seal
104		Not clean
105		Color not as specified
106		Not material specified
107		Individual pouches stick together or tear when separated
108		Pouch has foreign odor
	201	Tear notch missing
	202	Tear notch not located as specified
	203	Depth of tear notch not as specified

4.3.1.5 Unfilled pouch seal strength testing. The unfilled pouches shall be tested for seal strength in accordance with ASTM F 88. The lot size shall be expressed in pouches. The sample size shall be the number of pouches indicated by inspection level S-1. Three adjacent specimens shall be cut from each of the three sealed sides of each pouch in the sample. The results shall be reported to the nearest 0.1 pound. The average seal strength of each seal shall be calculated by averaging the strengths of the three test specimens cut from that seal. Any test specimen failing to meet the individual test specimen seal strength requirement specified in 3.1.5.2 or any seal failing to meet the average seal strength requirement specified in 3.1.5.2 shall be cause for rejection of the lot.

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4.3.1.6 Unfilled pouch integrity and air tightness of seals testing. The unfilled pouches shall be tested for conformance to the pouch integrity and air tightness of seals requirement in 3.1.5.2 by placing a rigid material, similar in size and shape to the product, in the appropriate size pouch (see table I) and sealing under the applicable vacuum requirement specified in table II. The pouch shall be examined for vacuum loss in accordance with the procedure specified in 4.3.2 not less than 96 hours after sealing. Any evidence of loss of vacuum shall be considered a major defect. The lot size shall be expressed in pouches. The sample unit shall be one unfilled pouch. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65.

4.3.2 Pouch vacuum examination. The filled and vacuum sealed pouches shall be visually examined for conformance to the vacuum requirement in 3.2 not less than 96 hours after filling and sealing. With the exception of fig bar pouch, the sealed pouch shall continue to exhibit tight adherence to the surface contours of the contents when a pulling force is applied at the center of each side seal. This force shall be applied by holding each side seal between thumb and forefinger of each hand, while simultaneously exerting a slight pull with both hands. Any evidence of loss of vacuum shall be considered a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65.

4.3.3 Filled and sealed pouch examination. The filled and sealed pouches shall be examined for the defects listed in table V. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65.

TABLE V. Filled and sealed pouch defects 1/

Category	Defect
Major	
101	Not clean
102	Tear, hole, or open seal
103	Evidence of delamination 2/
104	Closure seal not produced by specified method 3/
105	Closure seal width not as specified
106	Closure seal extends into or below tear notch
107	Pouch has foreign odor
108	Required marking missing, incorrect, or illegible
109	Contents caught in closure seal 3/
110	An impression or design on the closure seal surface which conceals or impairs visual detection of seal defects 4/
111	Fold over wrinkles in closure seal

1/ Any evidence of rodent or insect infestation shall be cause for rejection of the lot.

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2/ Delamination shall be scored as a defect except delamination of outer ply when located in the seal area 1/16 inch or further from food product edge of seal. Pouches exhibiting this type of delamination shall be tested by manually flexing the delaminated are ten times. The area of delamination shall be held between the thumb and forefinger of each hand with both thumbs and forefingers touching each other. The delaminated area shall be rapidly flexed by rotating both hands in alternating clockwise-counterclockwise directions. Care shall be exercised when flexing delaminated area near the tear notches to avoid tearing the pouch material. After flexing, the separated outer ply shall be grasped between the thumb and forefinger and gently lifted toward the food product edge of the seal. If the separated area is too small to be held between thumb and forefinger, a number two stylus shall be inserted into the delaminated area and a gentle lifting force applied against the outer ply. If separation of the outer ply can be made to extend to less than 1/16 inch from the product edge of the seal with no discernible resistance to the gentle lifting, the pouch shall be rejected.

3/ The effective closure seal is defined as any uncontaminated, fusion bonded, continuous path, minimum 1/16 inch wide, from side seal to side seal that produces a hermetically sealed pouch.

4/ If doubt exists as to whether or not the sealing equipment leaves an impression or design on the closure seal surface that could conceal or impair visual detection of seal defects, samples shall be furnished to the contracting officer for a determination as to acceptability.

4.3.4 Pouch closure seal testing. The filled and sealed pouches shall be tested for closure seal strength in accordance with ASTM F 88. Three adjacent specimens, 1/2 to 1 inch wide, shall be cut from the top closure seal of each pouch in the sample. The results shall be reported to the nearest 0.1 pound per linear inch. The average seal strength of the closure seal shall be calculated by averaging the test results of the three test specimens cut from the seal. Any test specimen failing to meet the individual test specimen seal strength requirement of 3.2 or any closure seal failing to meet the average seal strength requirement specified in 3.2 shall be scored as a defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.3.5 Shipping container examination. Examination of the filled and closed shipping containers shall be in accordance with the examination criteria contained in the appendix of PPP-B-636 and with the following defects added to the table: Major: (1) Marking missing, incorrect, or illegible, (2) Cushioning material missing or not as specified; Minor: (1) More than 30 pounds packed in box, (2) Bags not packed flat in layers.

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5. PACKAGING

5.1 Packing for shipment to assembler. Not more than 30 pounds of packaged product shall be packed flat in a fiberboard box conforming to style RSC-L, class domestic, variety SW, grade 200 of PPP-B-636. Cushioning material not less than 1/8-inch thick, conforming to type II, class B of PPP-C-843 or cushioning material not less than 1/6-inch thick, conforming to PPP-C-1797 or cushioning material conforming to type 7, class 1 of PPP-C-1752 shall be placed between each of the layers and in the bottom and top of the box to minimize movement of the individual packages.

5.2 Marking. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the title, number, and date of this document.

6.2 Use information. This document covers the packaging requirements for the following component items of the ration, cold weather:

Document No.	Product
	Entree component
MIL-C-44148	Chicken stew, cooked, dehydrated-(operational ration component)
MIL-B-44140	Beef stew, cooked, dehydrated-(operational ration component)
MIL-E-44141	Escalloped potatoes and pork, cooked, dehydrated-(operational ration component)
MIL-C-44137	Chicken a la king, cooked, dehydrated-(operational ration component)
MIL-C-44138	Chicken and rice, cooked, dehydrated-(operational ration component)
MIL-V-44139	Vermicelli with meat and sauce, cooked, dehydrated-(operational ration component)
	Other components
MIL-O-44136	Oatmeal cookie and granola bars-(operational ration component)
MIL-B-44135	Beverage base: orange compressed-(operational ration component)
MIL-C-44072	Cookies, oatmeal; and brownies, chocolate covered (for meal ready-to-eat)
MIL-F-44279	Fruit Bars (Fig and Blueberry Newtons)
MIL-T-44284	Tea, instant, sweetened, lemon flavored (operational ration component)

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Document No.	Product
	Other components (cont'd)
MIL-N-44280	Nut, raisin mix
MIL-S-44283	Soup, instant, reduced sodium chicken noodle
MIL-C-3031	Cocoa beverage powder, type 1, class 1
MIL-F-44282	Fruit soup (operational ration component)
MIL-B-44281	Beverage base, powder, apple cider, natural and artificial flavored (operational ration component)
MIL-O-44278	Oatmeal, instant (operational ration component)
MIL-C-10928	Candy and chocolate confections, type VII

6.3 Closure device. The Flip-Clip Fastener is manufactured by the Dennison Manufacturing Company, Framingham, MA 10701.

6.4 Subject term (key word) listing.

Cold weather rations
Packaging

Custodian:

Army - GL

Review activities:

Army - MD, TS
Navy - MC, SA
Air Force - 50
DP - SS

Preparing activity:

Army - GL

Project No. 89GP-0123

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-44276(GL)	Packaging of Ration, Cold Weather Components (In Pouches)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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